BIAS BOUNTY MAPPING EQUITY CHALLENGE · BEST DOCUMENTATION ENTRY

Coverage-gap scores without labels

A measured, reproducible pipeline for a score whose reference layers had been withdrawn

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0.058710
private RMSE of the final blend f2, rank 36, with no label and no withdrawn layer

81
files submitted, each with its purpose, expected score and SHA-256 logged before upload

r = 0.987
agreement of 73 signals' correlations measured on the public 30 % and the private 70 %

61 / 61
signal files of f2 rebuilt bit for bit from the raw data, in one 49-minute run

On 25 September the four reference layers left the challenge package, and with them every way to compute the target. We kept the organiser's formula and replaced each unknown reference count with a random variable built from permitted data (Section 4). The public leaderboard then served as a measuring instrument: two constant files gave the mean and spread of the hidden truth, and every later file gave one covariance with it (Section 6). The final file combines 60 such measurements in closed form. They held on the private set (Section 8), and one command rebuilds the final file bit for bit from the raw data (Section 11).

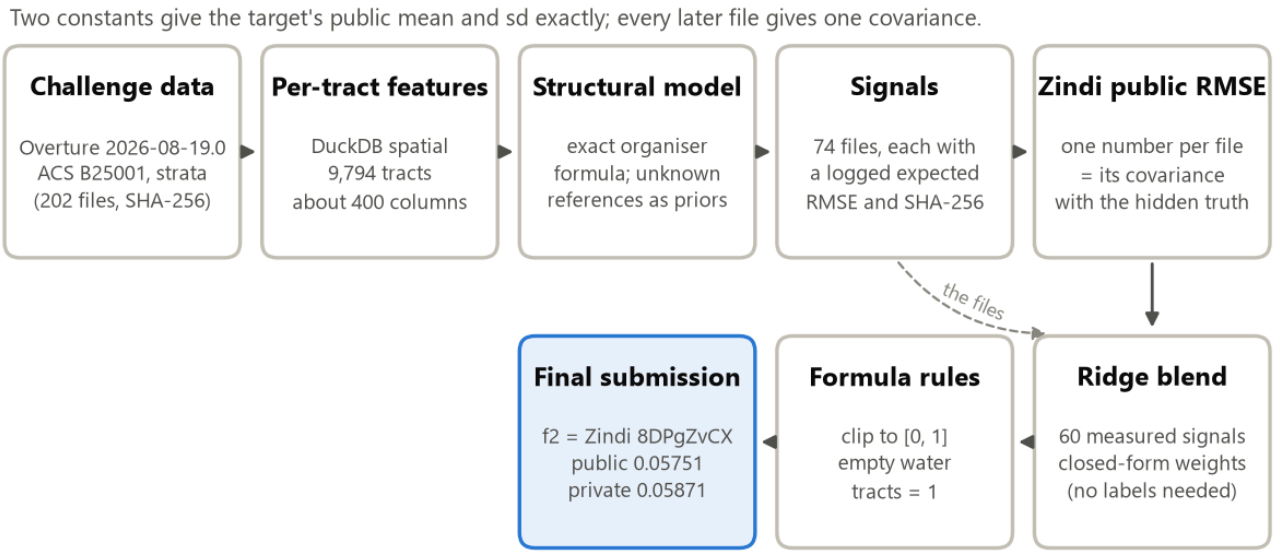


Figure 1. The pipeline. Every box except "Zindi public RMSE" runs locally and is deterministic; Zindi only returns one public RMSE per uploaded file.

Associated submissions: 8DPgZvCX, final blend f2 (public 0.057508, private 0.058710, rank 36 on the private leaderboard), and **EboKdJHP**, s111, second selection. Every other submission ID is in Appendix A.
Code, data checksums and every submitted file: https://github.com/Arnaud-TOE/bias-bounty-coverage-gap

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Where to find what the Info page asks for

What the Info page asks for	Where
Data sources	Section 2
How each component was computed	Sections 3 and 4
Edge cases: zero-population, water-dominated and no-Overture tracts (and others)	Section 5
Alternative weightings and why	Section 7
Reflection on the Bias Scorecard	Section 9
Reproduction	Section 11: `python scripts/reproduce.py` rebuilds the final file bit for bit; `--full` rebuilds everything from the raw data

Three ways to read this document

- **In ten minutes:** the first page, the Summary, Figures 1–3 and Section 8.
- **To audit the rules:** Sections 1 and 10, then Appendix A: every file, its purpose and its expected score, logged before upload.
- **To reproduce:** Section 11. One command rebuilds f2 in about ten seconds; a second rebuilds everything from the raw data in 49 minutes.

0 Summary

1. **There are no labels.** The reference layers (TIGER roads, Microsoft footprints, USGS/HIFLD facilities, County Business Patterns) were withdrawn and rebuilding the target with them is forbidden. Nothing can be trained in the usual sense.
2. **We modelled the organiser formula itself.** For every tract and component, the Overture count O is observed exactly. The reference count R is unknown, so it is drawn from a distribution built only from permitted data. The exact formula `1 − min(1, O/R)`, with its undefined cases, is averaged over 1,000 Monte-Carlo draws (Section 4).
3. **We used the public leaderboard as a measuring instrument, not as an oracle.** Two constant submissions gave the public target mean and standard deviation exactly. After that, every submitted file gave one number: its covariance with the hidden truth. We submitted 74 such measurements (81 files in all, with the constants, one probe and the blends). Before each upload we logged an expected RMSE and the file's SHA-256 (Section 6).
4. **The final file is a closed-form, ridge-regularised linear blend of 60 measured signals.** The weights are `w = (C + λ·diag C)⁻¹ c`. No tract-level value was ever inferred, and no group smaller than 300 scored tracts was ever probed.
5. **The method held out of sample.** For 73 signals, the correlation with the truth implied by the public 30 % and the one implied by the private 70 % agree with r = 0.987 (Figure 2).
6. **Everything reruns.** `python scripts/reproduce.py` rebuilds f2 bit for bit in about ten seconds. `python scripts/reproduce.py --full DIR` rebuilds every feature and every signal from the raw challenge data in a fresh environment. On 1 November all 61 signal files and f2 itself came out bit for bit identical to the files Zindi scored (Section 11).

1 The problem as it stood, and the rules we set ourselves

The target is the organiser's composite coverage-gap score, between 0 and 1, for 9,794 Census tracts. It is the mean of the **defined** components among three:

- **transport:** Overture motorway/trunk/primary/secondary km against TIGER S1100+S1200 km;
- **building:** Overture footprints against Microsoft footprints;
- **POI:** the mean of two halves:
  - a facilities half, which is the mean of the fire, EMS and school gaps against USGS/HIFLD;
  - an establishments half: all Overture places against CBP establishments.

Each gap is `1 − min(1, O/R)` and is undefined when R = 0. On 25 September the four reference layers left the package and reconstructing the target became a rule breach. On 8 October the staff confirmed that no tract-level label exists. The only feedback left was one public RMSE per submission, computed on about 30 % of the scored tracts.

This shaped every choice. We wrote the following rules into `docs/decisions.md` before using the leaderboard. They are enforced in code where possible.

Id	Rule	How it is enforced
C1	Never read a withdrawn layer, or a copy of one. Never use other participants' predictions, the scorecards of near-zero RMSE rows, or published group means.	`src/bbx/io.py` refuses any path matching `tiger`, `microsoft`, `hifld`, `usgs-structures`, `cbp`, `coverage-gap` … (unit test `tests/test_guard.py`)

Id	Rule	How it is enforced
D2	Never use strata columns derived from building footprints: USFS building, housing-unit and population layers (built on Microsoft footprints), and `carbonplan_buildings`.	The same module drops them on read (`drop_forbidden_columns`).
D1	Use the leaderboard only to measure global quantities. The cap rose from 20 to 74 measured signals over the month, with the user's agreement at each step. Signals must be continuous or cover a group of at least 300 scored tracts, all weights are ridge-regularised, and nothing is probed at the tract level.	Every file's purpose and expected RMSE are logged before upload (`submissions/log.csv`). The cap is checked in `scripts/make_finals.py`.
—	External data only for the Bias Discovery prize, never imported by `src/`.	`discovery/` is a separate tree.

2 Data sources

All data come from the public Source Cooperative bucket `s3://us-west-2.opendata.source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge`. They were retrieved on 16 October 2026 with `aws s3 sync --no-sign-request`. All **202 files** are checked against the bucket listing (byte counts, `data/bucket_listing.txt`) and a SHA-256 manifest (`data/MANIFEST.sha256`) by `scripts/verify_download.py`.

Dataset	File pattern	Version	Used for
Overture buildings	`reference/<region>/<region>-overture-buildings.parquet`	release 2026-08-19.0	building counts, sources, sizes, OSM names and classes
Overture roads (drivable classes)	`…-overture-roads.parquet`	2026-08-19.0	scored-class km O, route networks (`routes[].network`), names
Overture roads (unfiltered), rail, infrastructure	`…-overture-roads-unfiltered.parquet` etc.	2026-08-19.0	service roads, tracks, road history
Overture places	`…-overture-pois.parquet`	2026-08-19.0	fire, EMS and school counts (organiser category lists), all places, alternate categories
ACS B25001 housing units	`…-census-acs-housing.parquet`	ACS 2020–2024	housing units
Strata table (232 columns)	`strata/<region>/<region>-strata-tract-table.parquet`	per-source vintages	population (urban/rural), RUCA, SVI, CVI, tribal overlap, hazards
Tract polygons	`strata/<region>/<region>-census-tracts.parquet`	TIGER 2025 cartography, 2020 tracts	spatial joins, land area (ALAND)
Scored tract list	`reference/<region>/<region>-sample-submission.csv`	—	9,794 GEOIDs: TX 6,003 · AZ 1,593 · OK 1,192 · CA 591 · WA 415

Never read by the scoring pipeline:

- TIGER roads, Microsoft footprints, HIFLD/USGS facilities, CBP, and any equivalent copy;
- the footprint-derived strata columns (`usfs_Building*`, `usfs_HU*`, `usfs_Pop*`, `carbonplan_buildings`);
- bookkeeping strata columns (vintages, labels, names), which carry no signal.

The organiser category lists are copied verbatim into `src/bbx/config.py`:

- fire: `fire_department`;
- EMS: `ambulance_and_ems_services`;
- schools: six categories, `elementary_school` … `public_school`.

Public knowledge we used, with credit. Some calculation conventions of the reference come from methods that participants published on the Zindi discussion board. We used the conventions as knowledge. We never used their code, their files or the withdrawn layers.

- *PeterTheDataScientist*: the reference clips TIGER lines to tracts in EPSG:5070 and keeps one record per route number.
- *HHYurdagul* (#35478): a highway running along a tract boundary is credited to both tracts.
- *iam-dante*: TIGER keeps a duplicate record for a route's local name.
- *Luca925* (#34925): a pattern for fire stations filed under other categories.

The TIGER technical documentation (2024 edition, section 4.12.4) confirms that the All Roads files hold one spatially coincident segment per road name.

3 Per-tract features

All features are built with DuckDB spatial (`src/bbx/features*.py`), one row per scored tract. The modelling table has 9,794 rows × 267 columns, plus 147 columns in eight extra feature files built in later passes.

Geometry conventions, unit-tested in `tests/test_data_contract.py`:

- The layers are lon/lat (OGC:CRS84). Spheroid lengths use `ST_Length_Spheroid(ST_FlipCoordinates(geom))`. Without the flip, the EPSG:4326 axis order silently returns NaN. The test checks a known motorway against its EPSG:5070 length (difference under 1 %).
- Planar measurements are in EPSG:5070, the organiser's projection: clipped km, distances to the boundary, buffers.
- GEOID is always text. 2,183 GEOIDs start with 0 (Arizona 04, California 06).
- All areas use **ALAND** (land only), with a 0.01 km^2 floor, so water never dilutes a density.

Group	Main features (per tract)	Module	Check
Roads	km of each Overture class clipped to the tract; km by **route level** (highest `routes[].network`: Interstate, US, state, Texas FM/RM/Loop/Spur, county, other); unnamed km; km last edited before 2012	`features.py`	79,164 km clipped against 79,064 km in the layer (+0.13 %, boundary overlaps)
Road reference artefacts	route km within 10 m outside the tract (boundary band); route km weighted by the number of distinct route designations (multiplicity); local-name km; long boundary runs (> 232 m in a 30 m ring)	`features_roadref.py`, `features_bandruns.py`, `scripts/build_uncounted_majors.py`	ratios against the published 0.71–1.59 range (Section 4.1)

Group	Main features (per tract)	Module	Check
Buildings	count by centroid, first source (OSM, Microsoft ML, Esri, USGS), multi-source, footprints under 50 m^2, area; hand-drawn footprints absorbing Microsoft ones	`features.py`, `features_bldref.py`	1,081,345 of 1,081,346 eastern-wa footprints assigned
Places	organiser category counts; fire stations filed under other categories (name pattern, vendors excluded); alternate categories; closed places; community anchors; facilities within 300 m / 1 km across the boundary; postcodes	`features.py`, `features_extra.py`, `features_osmfac.py`	category lists identical to the README
Facility evidence in buildings	OSM fire-station, EMS and school buildings (Overture buildings carry OSM names and classes; Overture places do not)	`features_osmfac.py`	—
Business proxies	large-building counts and areas, distinct place addresses, dominant postcode	`features_bizaddr.py`	—
Strata	population (urban/rural), RUCA, SVI, CVI, tribal overlap, wildfire, heat, drought	`dataset.py`	footprint-derived columns dropped

Totals over the 9,794 tracts:

- 19,169,788 buildings;
- 2,001,075 places;
- 192,520 km of scored-class roads;
- 1,069,751 km of drivable roads.

4 How each component was computed

4.0 Principle: the expected value of the exact formula

For tract t and component k, the organiser computes `g = 1 − min(1, O/R)`. It is undefined when R = 0, and the score is the mean of the defined components. O is observed exactly from Overture. R is replaced by a random variable whose law uses permitted data only. The prediction is the **Monte-Carlo expectation of the exact formula**, undefined cases included:

```
score(t) = E[ mean over defined k of (1 − min(1, O_k / R_k)) ],   1,000 draws, seed 20261016
```

The undefined cases matter: when one component is undefined, the score is the mean of the other two, so the divisor changes from tract to tract. This expectation is the RMSE-optimal point prediction under the stated beliefs. `src/bbx/structural.py` also returns each component's **exact additive contribution** to the expected score: `c_k = E[1(k defined) · g_k / n_defined]`, so the contributions sum to the score (unit-checked). These contributions are what we submit as signals: each one measures how well one mechanism tracks the truth.

4.1 Transport: the decisive component

Removing the 17 transport signals would add +0.0122 to the expected RMSE of the final blend (Figure 5).

O, the Overture side. Km of `motorway`, `trunk`, `primary` and `secondary`, clipped to the tract polygon in EPSG:5070. This is the organiser's own definition, so it is exact.

R, the TIGER S1100+S1200 length, is not observable. We model it as

```
R = proxy × exp(0.5 · ε),   ε ~ N(0, 1)
proxy = major km × multiplicity
      + boundary credit
      + w_FM · Texas Farm/Ranch-to-Market km
      + w_cty(region) · county-route km
      + w_loc · local-name copies
```

The pieces are as follows.

- **Major km.** The route-tagged km of Interstate, US and state highways. They also include the state loop, turnpike and toll systems that TIGER files as primary or secondary roads: Phoenix Loops 101/202/303, the Oklahoma turnpikes, and the Texas Loops, Spurs, Beltways and toll roads.
- **Multiplicity.** TIGER keeps one record per route number, and the reference keeps every copy. A segment carrying I-35 and US-81 therefore counts twice.
- **Boundary credit.** Tract boundaries follow TIGER road edges, so a highway that forms a boundary is credited in full to both tracts. Overture's copy, a few metres to one side, lands in only one of them. We measure route km within 10 m outside the tract (v3, v7), or long runs of more than 232 m within a 30 m outer ring (v9).
- **Calibration of the two free weights.** `w_FM` and `w_cty` are solved from the organisers' published figures only. The Info page gives the Overture/TIGER length ratio as 0.71 to 1.59 across the four regions. Texas sits at the top of the range and Oklahoma at the bottom; Arizona and California fall inside it without being fitted (1.32 and 1.43 under v7L).
- **Definedness.** The transport component is defined when R > 0. Its probability is:
  - 1 with any major route;
  - 0.19 when the only routes are Texas FM/RM roads;
  - by region when the only routes are county routes: Oklahoma 1, Arizona 0, California 0;
  - 0.03 with no route at all.

  These values reproduce the published shares of tracts with at least one undefined component:

Region	Published	Model v7
Eastern Oklahoma	21 %	21.1 %
Northern California	37 %	37.9 %
South-Central Texas	28 %	26.8 %
Maricopa	55 %	52.5 %

Successive versions. Each version was measured by submitting its transport contribution alone. ρ is the implied correlation with the truth, on the public and private sets.

Version	Mechanism added	Signal	Public RMSE	ρ public / private
v1	route-level proxy, definedness calibrated to the published shares	s06	0.08210	0.42 / 0.40
v3	boundary credit (10 m band) and route multiplicity	s77	0.07426	0.56 / 0.57

Version	Mechanism added	Signal	Public RMSE	ρ public / private
v7	route classification fixed: Phoenix Loops and Oklahoma turnpikes are highways; Texas FM/RM nearly absent from TIGER (w_FM 0.112, w_cty OK 0.152)	**s97**	**0.06648**	**0.67 / 0.70**
v9	boundary credit from long runs in a 30 m ring (w_FM 0.005, w_cty 0.095)	s106	0.06902	0.64 / 0.66
v7L	local street names as one more TIGER record (weight 0.25; w_FM 0.051, w_cty 0.075)	s140 (v7L − v7)	0.08630	0.35 / 0.40
v7LF	local names of Interstates and Texas loops as well	s146 (delta)	0.08851	0.14 / 0.06
v7Linv	the published ratio read the other way round (TIGER/Overture): local names 0.35, w_FM 0.175, w_cty 0.235	s147 (delta)	0.08578	0.40 / 0.37

> **Key finding:** the v3 → v7 step was our largest single gain (public 0.0743 → 0.0665). It came from reading Overture's `routes[].network` correctly: `US:AZ:Loop` and `US:OK:Turnpike` are freeway systems, not county roads. It also came from recognising that the 50,600 km of Texas FM/RM roads are almost all S1400 in TIGER.

The deltas (v7L, v7LF, v7Linv) are submitted as differences from their parent, with the same random draws. This lets the blend weigh each mechanism separately.

4.2 Building

- **Model.** `R = round(O × exp(−0.05 + 0.15·ε))`. The README says Overture usually holds at least as many footprints as the reference, so the expected building gap is small everywhere.
- **What the leaderboard said.** The building family adds only 0.0002 to the final blend.
- **One informative proxy.** The best building signal is the **share of Overture buildings sourced from Microsoft ML** (s13, ρ = 0.29 / 0.30). Where local mapping is thin, the other components have gaps too.
- **Other signals measured:**
  - footprints under 50 m^2 (s27);
  - Esri-sourced share (s28);
  - buildings per ACS housing unit (s26, negative);
  - hand-drawn footprints that absorb several Microsoft ones (s80).

4.3 POI: fire stations, EMS, schools and establishments

Facilities half. For each type, the reference count is H = M + K:

- M ~ Binomial(O, π) are the Overture records that are real reference facilities;
- K ~ Poisson(max(λ − π·O, 0)) are the facilities the map misses.

Type	π	λ (expected reference facilities)	Extra evidence that forces H ≥ 1
Fire (v1)	0.8	rural pop/4,000 + urban pop/18,000 + km^2/2,500 (national aggregates, about 29,700 US stations)	a station name filed under another category: P = 0.85
Fire (v8)	0.8	same	name-verified misfiled station (vendors excluded), OSM fire-station building, `fire_department` as alternate category: P = 0.9

Type	π	λ (expected reference facilities)	Extra evidence that forces H ≥ 1
Fire (v9)	0.8	rural pop/**2,000** + urban pop/**36,000** + km²/2,500	v8 evidence plus **town cores** (RUCA 4 and 7): P = 0.9
EMS	0.9	0.9·O + 0.1·Ô (Ô: Poisson GLM smoothing of Overture counts on population, area, rurality, region)	—
Schools	0.9	0.9·O + 0.15·Ô	—

The facilities half is the mean over the defined types.

The v8 and v9 changes were each measured before adoption:

Fire change	Signal	ρ public / private
Clean station evidence	s100	0.17 / 0.17
Town-core stations	s134	0.17 / 0.16
Recalibrated rural and urban rates	s136	0.28 / 0.29
Rural populated tract with no Overture station	s14	0.21 / 0.23

Rural volunteer stations are numerous and poorly covered by Overture places. City stations are fewer per resident and well covered.

Establishments half. The proxy is `R = round(√(max(0.0243·pop, 0.5) · max(places/2, 0.5)) × exp(0.5·ε))`, with 0.0243 establishments per resident nationally and about two Overture places per establishment. The organisers allocate ZIP-level CBP counts to tracts with a crosswalk of business addresses, which we may not use. We therefore submitted several **allocation hypotheses** as separate signals and let the leaderboard weigh them:

- by area (s24, s38);
- by population (s25);
- by housing units (s36);
- by ZIP spread (s42, s43);
- by large-building and business-address proxies (s86).

Individually these signals are weak, sometimes negative. Their contrasts carry +0.0020 in the blend.

4.4 The composite, and what we learned about it

The composite is the mean of the defined components, with the POI component itself a mean of its halves. The model reproduces the formula's undefined cases exactly, draw by draw.

Weakness 1: the level is off. The structural composite averages 0.095–0.107 on the four regions, depending on the version. The measured public mean is 0.0598, so the facility terms over-predict gaps.

Weakness 2: the formula's weights are not the best weights. The composite with the formula's own weights predicts worse than its transport part alone:

Model	Composite	Transport contribution alone
v1	ρ = 0.33 (s04)	0.42 (s06)
v3	ρ = 0.47 (s79)	0.56 (s77)

We therefore use the structural model as a **generator of mechanisms**: contributions, probabilities of definedness, and the edges of the score's distribution. The leaderboard weighs them (Sections 6–7). The probability that the score reaches 0.30 (s117, ρ = 0.38) is one such edge. A linear blend cannot reproduce the skewed tail of the score without it.

5 Edge cases

Counts are over the 9,794 rows, recomputed from the modelling table.

Case	Tracts	How it is handled
Zero population Info page	27 (67 with < 100 residents; 63 special-use 98xx tracts)	Kept and scored like the others. Per-capita priors use a floor, so their facility expectation rests on land area. `pct_urban` is missing (27 tracts) and falls back to the regional median where a model needs it. 25 of the 27 have buildings and 20 have scored roads, so no rule fixes their score.
Water-dominated Info page (water > 50 % of the polygon, polygon area against ALAND)	43 (40 in Texas: Gulf coast bays and large reservoirs; 171 tracts are over 25 % water)	Every area and density uses ALAND, never the polygon area, so water does not dilute densities. Otherwise they are modelled like any other tract.
Pure water, no Overture data at all Info page	2: 48039990000 and 48057990000 (ALAND = 0)	**Formula rule** (`src/bbx/postprocess.py`, unit-tested). A scored tract has at least one defined component, and with O = 0 every defined component equals 1. Their score is therefore exactly **1.0**, written into every final file.
...and their effect on measurement	the same 2	A signal extreme on those two tracts gets a measured covariance driven by whether they fall in the public 30 %. Since 25 October every new signal is set to the target mean m on them (`bbx.signals.write_signal`). The 46 older signals kept their values (and rebuild with them, `LEGACY_WATER_SCRIPTS`); s07 is 18–20 standard deviations above its mean there (Sections 7–8).
No scored-class Overture road	137	The transport component is defined only through the route-based reference probability. Its gap is 1 when defined.
No major route (transport likely undefined)	3,788	The probability of definedness is calibrated as in Section 4.1.
No Overture fire / school / EMS place	6,912 / 2,100 / 9,078	The gap comes from the facility priors (Section 4.3).
Zero housing units	82	Housing-unit ratios are NaN and replaced by the median when a signal is standardised.
Missing SVI / CVI	88 / 2	Median fallback, used only in signals and in the bias analysis.
GEOID as text	all; 2,183 with a leading zero	Read with `dtype=str` everywhere; the test checks 1,592 Arizona GEOIDs keep "04".
New Mexico tract in the Maricopa package	35023970000	Treated as part of maricopa-az, as the Info page says.

Case	Tracts	How it is handled
Eastern Washington	415	The Info page says that all scored tracts are in the other four regions. Probe s124 (1.0 on these tracts, m elsewhere) scored exactly the public sd: **not public**. The finals therefore fit their weights on the four regions and carry them to Washington without region-specific signals. After the close, the private scores showed that **all 415 tracts are in the private set** (Section 8).
Submission format	—	Exactly `GEOID,coverage_gap_score` in the official row order. The grader reads the first column after GEOID (#35361), and the official sample puts `transport_gap` there. Values are clipped to [0, 1] (clipping can only lower the RMSE), with 6 decimals.
Unscored 99xx tracts in the package	7	Never used.

6 From mechanisms to a submission: the leaderboard as an instrument

6.1 The algebra

For a file p scored on the public subset, where the target has mean m and standard deviation s:

```
RMSE² = Var(p) + s² − 2·Cov(p, y) + (mean p − m)²
```

- Two constants, 0 and 0.10 (s01, s02), give **m = 0.059806** and **s = 0.089022** exactly (`bbx.lb_algebra.moments_from_constants`).
- Every later file then gives `Cov(p, y)`, and hence its correlation with the truth. The moments of p are taken over all tracts of the scored regions, as a proxy for the unknown public subset.

6.2 Signals

A mechanism x is submitted as `p = m + k·s·z(x)`, clipped to [0, 1], where z is the standardised mechanism and k is a modest scale (`bbx.signals.write_signal`). Each file is therefore a sensible prediction on its own. Before upload, `bbx.submit.write` validates the file and logs it:

- 9,794 unique GEOIDs, no blanks, values in [0, 1];
- its SHA-256, mean, sd, purpose and expected public RMSE are appended to `submissions/log.csv`.

After upload, `bbx.submit.record` stores the Zindi ID and score. All 80 logged scores were checked against the Zindi Submissions tab, with no difference.

6.3 The blend

```
ŷ = m + Σᵢ wᵢ (pᵢ − mean pᵢ),    w = (C + λ·diag C)⁻¹ c,    expected RMSE = √(s² − 2w·c + w·C·w)
```

- C is the covariance of the signals over the four scored regions and c the vector of measured covariances.
- Eastern-wa tracts get a second blend without region-specific signals (`fit_without_wa` in `scripts/make_finals.py`).
- The formula rule (Section 5) and clipping are applied last.
- f2 uses **60 signals with λ = 0.005**. Its weights are stored in `submissions/final_f2_blend_ridge0005_with_s07_s22_weights.json` and listed in Appendix A. They

apply to the submitted files, so their size depends on each file's scale k.

6.4 Does the instrument work? Pre-registered blends against reality

Blend	Signals, ridge	Expected public (logged before upload)	Public	Private
s16 (first blend, 18 Oct)	8, λ = 0.05	0.077974	0.078204	0.079596
s111 (validation blend, 26 Oct)	45, λ = 0.005	0.059738	0.059990	0.061256
f2 (final, 31 Oct)	60, λ = 0.005	0.054996	**0.057508**	**0.058710**
f5 (sent after the close, not ranked)	58, λ = 0.02	0.056746	0.056982	0.058048

> **Key finding:** three of the four blends landed within 0.0003 of the logged expectation. The fourth, f2, missed by 0.0025; Section 8 explains why.

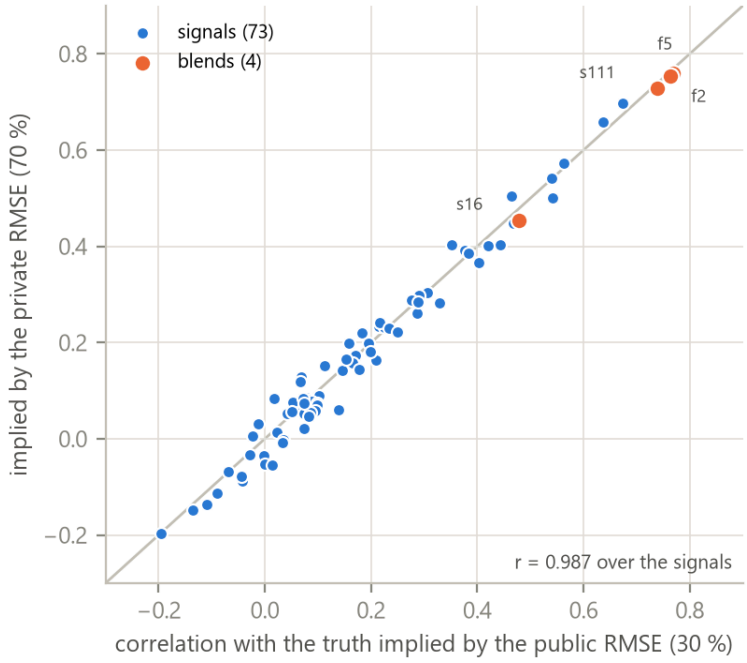


Figure 2. Each signal's correlation with the truth, measured twice: from its public RMSE (30 % of tracts) and from its private RMSE (70 % plus eastern-wa). The private moments come from the same two constants (m = 0.059915, s = 0.089342). Over 73 signals r = 0.987. The differences have sd 0.031, a little above the pure sampling noise of a correlation measured on 2,814 tracts (about 0.02), as expected with heavy-tailed signals. The measurements replicate out of sample.

6.5 Controlling over-fitting

1. **Few, global parameters.** One weight per measured signal and nothing at the tract level.
2. **Continuous signals or large groups.** Every regional or metro copy covers at least 300 scored tracts.
3. **Ridge on every weight.** λ was chosen by an end-to-end simulation (`docs/05_plan_audit.md`, check C2):
   - draw synthetic truths around the blend;
   - draw a random public 30 %;
   - recompute every implied covariance from simulated RMSEs;
   - refit the blend and score it on the synthetic private 70 %.

   The simulation recovered the covariances to within 1.4 % and favoured λ between 0.001 and 0.005.
4. **WEAK rule.** A recent signal whose removal raises the expected public RMSE by less than 0.00003 is left out of the finals. Thirteen signals are left out this way, among them the four metro copies s125–s128.

7 Alternative weightings, and why we chose what we chose

Alternative	What it means	Evidence	Decision
The formula's own component weights (equal mean of defined components)	Submit the structural composite	Composite ρ 0.33 (v1) and 0.47 (v3), below its own transport part (0.42, 0.56)	Rejected the leaderboard weighs mechanisms. On 26 October the blend gave the v7 transport contribution about 1.25 times its model weight.
Component level against contribution	Submit the transport gap g itself, not its share g/n_defined of the score	v7 gap s101 ρ 0.47, against v7 contribution s97 ρ 0.67	Kept the contribution, which carries the formula's divisor.
Population weighting	Weight tracts by residents	The metric is an unweighted RMSE over tracts	Rejected the metric weighs every tract equally.
One weight per region	Region-restricted copies of transport and fire	Oklahoma transport copy s83: ρ 0.44 alone and the largest marginal gain of its day. Arizona, California and Texas copies: smaller gains	Kept in the blend. The weights are not used for eastern-wa, which is absent from the public set.
One weight per metro area	Houston, Phoenix, San Antonio and Austin copies (s125–s128)	−0.00010 in total	Dropped by the WEAK rule.
Allocation of CBP establishments	By area, population, housing units, ZIP spread or business proxies	Each alone ρ between −0.11 and +0.10; together +0.0020 in the blend	All kept each one measured and weighed by the blend.
Ridge strength	λ = 0.005 (f2) against 0.01 (f3) and 0.02 (f5)	Simulation: λ 0.001–0.005 best. Reality: f5 (λ 0.02, without s07/s22) private 0.058048, better than f2's 0.058710	λ = 0.005 chosen on the simulation for f2. In hindsight more shrinkage was better, but the comparison is confounded with s07/s22 (next row).
With or without s07/s22 (two signals driven by the water tracts)	f1/f3/f5 drop them; f2 keeps them	Their covariance depends on whether the two water tracts are public. s07's implied correlation flips sign from public (+0.013) to private (−0.054)	Hedged f2 kept them, as one side of a hedged pair. The private result favours dropping them (f5).
Ridge towards the structural composite instead of towards zero (f4)	Shrink the weights towards those that reproduce structural v9L	Formula: slight gain. End-to-end simulation: worse in 95 % of synthetic truths	Rejected the simulation outweighs the formula.
Pruning weak signals	WEAK rule against keeping every signal (f1)	Simulation: if those signals are noise, keeping them costs 0.0002 of private RMSE; if they are real, dropping them costs nothing	Applied to f2, f3 and f5.
Hedged selection	Two finals that differ where we are unsure	Zindi keeps the better of the two selected files on the private set	Hedged f2 + s111 selected, and checked on every page because of a Zindi selection bug (thread #35607).

8 What the private leaderboard taught us

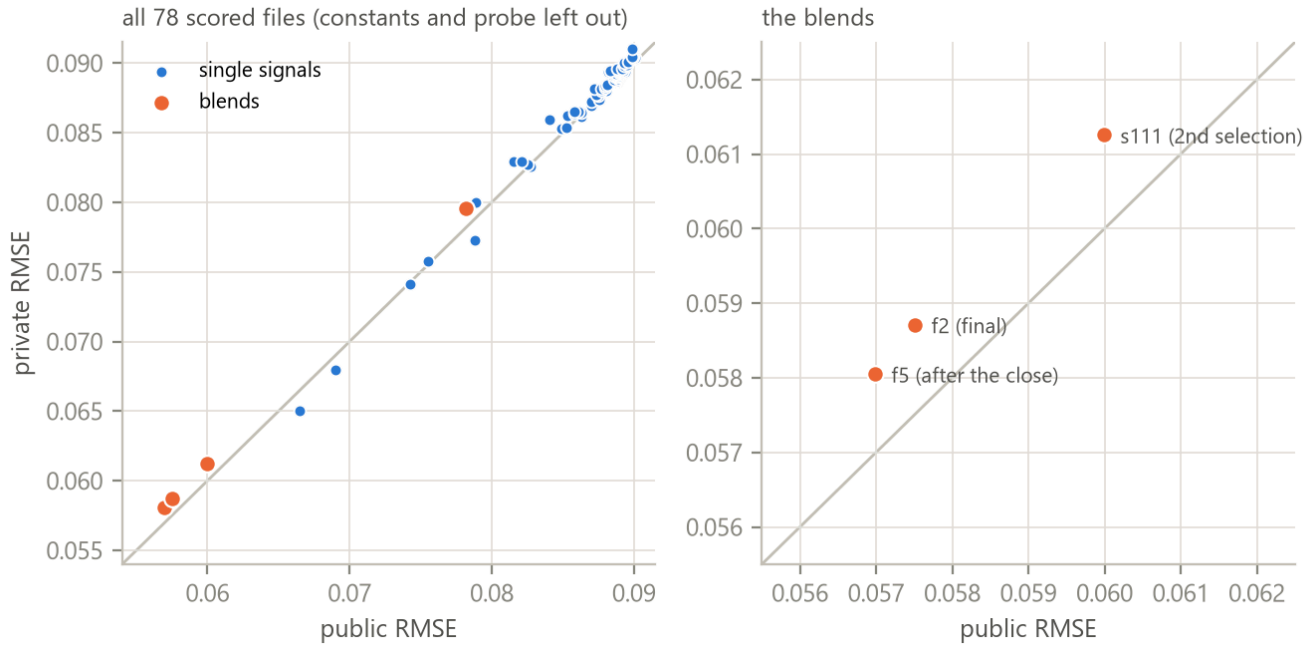


Figure 3. Public against private RMSE for every scored file. The ranking of files is preserved (rank correlation 0.94). Single signals lose 0.0003 on average from public to private; every blend loses 0.0011 to 0.0014.

1. **The measurements replicate** (Figure 2). Correlations measured on 2,814 public tracts predict those on the private 70 % with r = 0.987.
2. **Blends lose about 0.0012 from public to private, consistently** (s16 +0.0014, s111 +0.0013, f2 +0.0012, f5 +0.0011). This is the expected cost of weights fitted on noisy public covariances, plus eastern-wa (point 4).
3. **f2's public miss had one cause.** f2 scored 0.0025 above its expectation; f5 missed by only 0.0002.
   - For a linear blend $B = a + \Sigma\, w_i\, p_i$, the public MSE is an exact, truth-free function of the signals' public RMSEs, except for one term: $\text{mean_public}(B^2 - \Sigma\, w_i\, p_i^2)$, which depends on which tracts are public.
   - In f2 the two empty water tracts (fixed at 1 by the formula rule) carry values of that term of 0.70 and 0.66. No other tract exceeds 0.01. The cause is s07, weighted +3.1 and 18–20 standard deviations above its mean on those two tracts.
   - Depending on whether zero, one or both of them are public, this term alone moves f2's public RMSE by about −0.0013, +0.0009 or +0.0030. The observed shift was +0.0025.
   - f5 drops s07 and sits on its expectation.

   > **Lesson:** a signal that is extreme on a handful of tracts makes its own measurement fragile. We neutralised new signals on those tracts on 25 October, but kept the old ones.
4. **Eastern Washington is in the private set.** Two private scores pin this down:
   - the probe s124: 1.0 on eastern-wa, m elsewhere, private RMSE 0.2385;
   - the indicator s31.

   Solved together, they give eastern-wa a share of 0.0595 of the private set. With 6,565 private tracts in the four regions (70 % of 9,379), that is exactly **415 tracts: all of eastern-wa**, with a mean target of 0.093. f2 predicted a mean of 0.076 there, with weights carried over from the other regions. The bias

costs about 0.0001 of private RMSE. It is small, but it contradicts the Info page, and it matters for anyone reusing the scorecard (Section 9).

5. **What we would do differently:**
    - neutralise every signal on the rule-determined tracts from the first day;
    - prefer the more shrunken final when two candidates are close;
    - measure the Overture/TIGER transport reference with fewer, bolder variants earlier. Transport carries most of the gain, and its uncertainty was our largest remaining error.

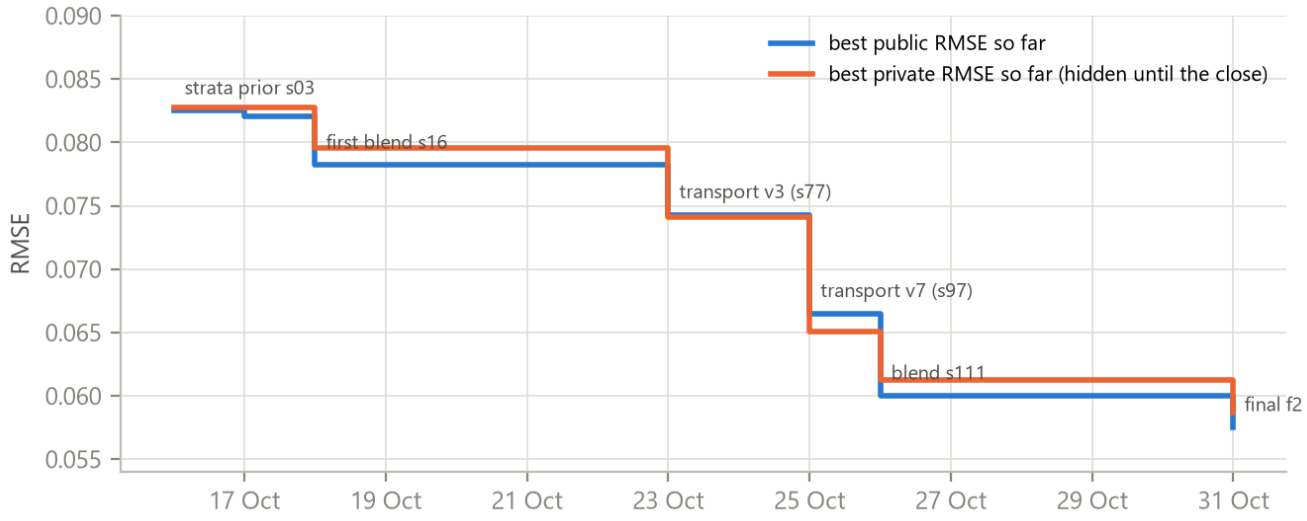


Figure 4. Best public RMSE over time and the best private RMSE among the same files (hidden until the close). The big steps are mechanisms of the transport reference (v3, v7); the blends then add 0.006–0.009.

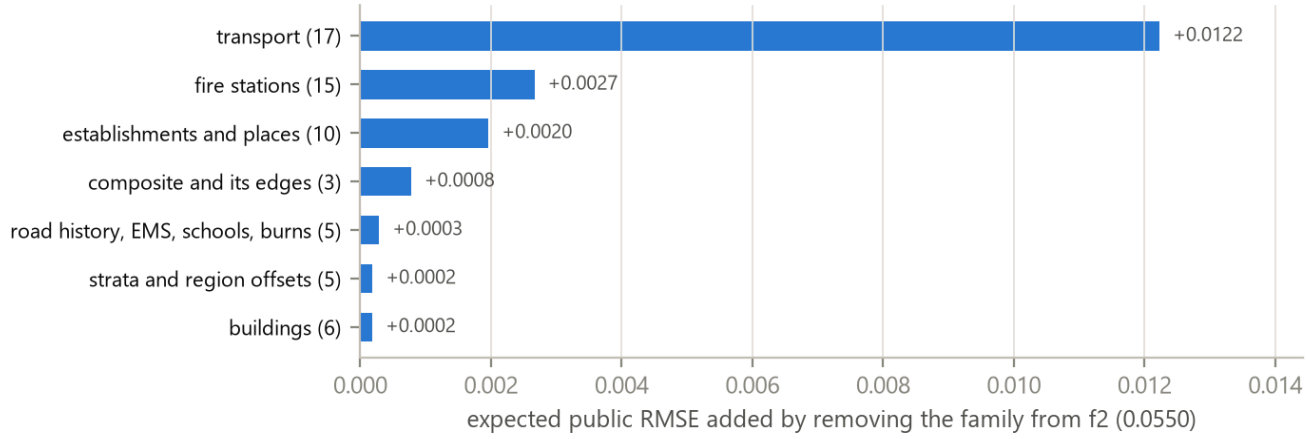


Figure 5. What each family of signals is worth inside f2: the increase of the expected public RMSE (0.0550) when the whole family is removed. Alone, the 17 transport signals would give 0.0615.

9 Reflection on the Bias Scorecard

What the scorecard measures for a label-free model. Zindi's scorecard of a submission is computed from **that submission's own predictions**, by stratum. We recomputed f2's values exactly from the file: the mean score by stratum over all 9,794 rows gives rural/urban **2.25×** and tribal **2.71×**, the figures Zindi shows. It carries no information about the truth. For a model like ours it describes the model's beliefs, so it is worth reading as a check on those beliefs.

Stratum (f2)	Disparity ratio
Rural / urban	2.25
Tribal / non-tribal	2.71
SVI top quartile / bottom quartile	1.27
CVI top / bottom quartile	1.59
Wildfire hazard (WHP) top / bottom quartile	2.03
Summer heat days top / bottom quartile	0.51
Summer drought (DSCI) top / bottom quartile	0.85

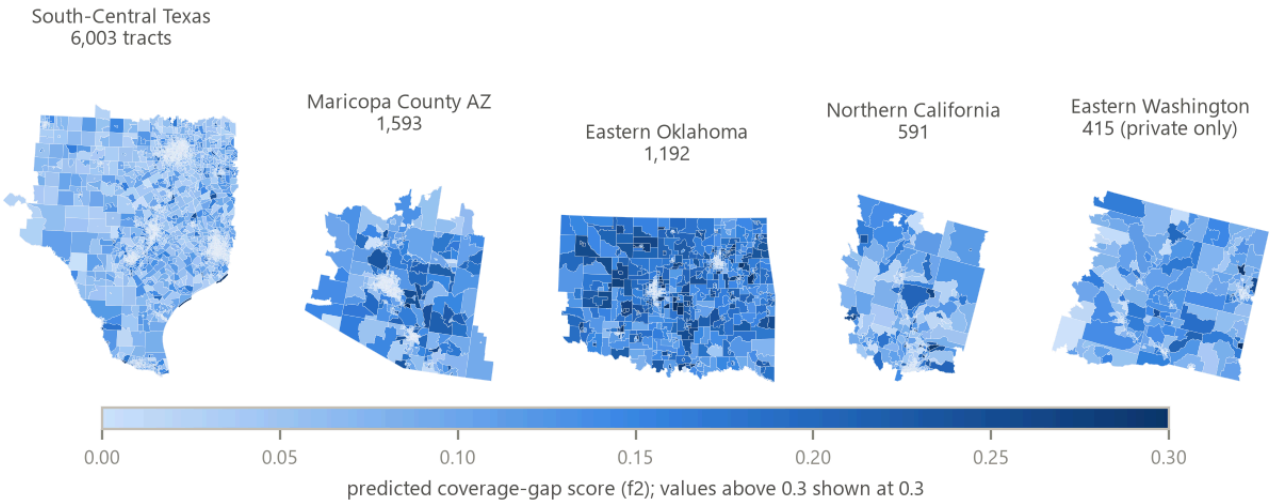


Figure 6. f2 predictions by tract. Gaps concentrate in rural Oklahoma, the outer counties of Maricopa and the northern California foothills. Urban cores are near zero.

Three observations about the scorecard:

1. **Pooling regions inflates the rural disparity.** Within each region, f2's rural/urban ratio is:

Region	Rural / urban, within the region
Texas	1.83
California	1.81
Oklahoma	1.88
Arizona	3.50

   Pooled, the ratio is 2.25. Rural tracts are over-represented in Oklahoma, the region with the highest gaps: 22 % of rural tracts, against 12 % of all tracts. The Info page itself warns that transport gaps "are not comparable across regions", because the Overture/TIGER ratio runs from 0.71 to 1.59. Part of a pooled rural disparity is a regional definition effect, not a mapping deficit. We suggest stratifying the scorecard by region.

2. **Good RMSE submissions understate extreme gaps.** An RMSE-optimal prediction shrinks towards the mean: f2's sd is 0.069, against 0.089 for the target. Apart from the two water tracts fixed at 1, no f2 tract exceeds 0.41. Threshold metrics such as the POI Desert Index (poi_gap ≥ 0.8) therefore report fewer deserts for better submissions. Group means are preserved only when the group is in the model's span, which is why we kept the strata prior s03 (rural and tribal shares) in every blend.

3. **The formula itself drives part of the rural and tribal disparity.** The truth combines a one-sided ratio (more Overture than reference counts as full coverage) with a variable divisor:
    - a rural tract with no named highway has no transport component;
    - its score is then the mean of building and POI;
    - its POI gap is often 1 for fire stations, because volunteer stations are poorly covered.

    In eastern Oklahoma, county roads and turnpikes decide whether transport is defined at all. Disparities in the scorecard thus mix mapping gaps with reference-definition effects. Our structural model separates them (the component contributions in Section 4), and we recommend that kind of decomposition to anyone using the score for emergency planning.

Finally, **eastern Washington** (415 tracts) is scored in private but appears nowhere in public, so no participant could calibrate it. Any scorecard pooled over the 9,794 rows includes predictions that were, by construction, extrapolated.

> **Recommendation:** stratify the scorecard by region, and decompose a disparity into the formula's components (Section 4) before using the score for emergency planning.

10 Compliance appendix

Topic	What we did
Never read	Withdrawn layers (TIGER, Microsoft footprints, HIFLD/USGS, CBP) or copies obtained elsewhere; footprint-derived strata columns; other participants' predictions, the scorecards of near-zero RMSE rows, and published group means. The I/O guard and its tests enforce the first two.
Leaderboard use	74 measured signals, the 2 constants, one region-level probe (s124) and 3 blends; f5 was sent after the close (81 files in all). No tract-level or sub-300-tract probing. Every file's purpose and expected RMSE were logged before upload.
External data	Only in `discovery/` (our Best Bias Discovery entry, "phantom homes"); never imported by `src/`.
Question to the organisers	On the footprint-derived strata columns: https://zindi.world/competitions/bias-bounty-mapping-equity-challenge/discussions/35402. No answer came, so the columns stayed excluded.
Tools	Python and open-source packages only; no AutoML, no paid API.
Coding assistant	A general-purpose coding assistant helped draft code and text. It is not needed to run or reproduce anything (rules: "General-purpose development assistants … may be used").
Licence	Challenge data are CC-BY-SA 4.0. Our derived files are shared under the same licence.

11 Reproduction

Environment:

- Windows 11 Pro, Intel Core i7-10610U laptop (4 cores), 32 GB RAM, no GPU;
- Python 3.13, duckdb 1.5.6, pandas 3.0.6, numpy 2.5.4, statsmodels 0.15.0, geopandas 1.2.0 (all pinned in `requirements.txt`);
- the structural model uses seed 20261016.

```
python -m venv .venv
.venv/Scripts/pip install -r requirements.txt        # .venv/bin/pip on Linux/macOS
.venv/Scripts/pip install -e .
B=s3://us-west-2.opendata.source.coop/humane-intelligence/bias-bounty-mapping-equity-challenge
python -m awscli s3 sync $B/reference/  data/raw/reference/  --no-sign-request
python -m awscli s3 sync $B/strata/     data/raw/strata/     --exclude "*.csv" --no-sign-request
python -m awscli s3 sync $B/boundaries/ data/raw/boundaries/ --no-sign-request
python scripts/verify_download.py       # 202 files: byte counts and SHA-256
python -m pytest -q                     # 13 tests: guard, algebra, data contract, formula rule
```

Depth 1: the final file from the archived signals (about 10 s).

```
python scripts/reproduce.py
```

The script:

- checks the SHA-256 of every signal file of f2 against `submissions/log.csv`;
- refits the blend exactly as `scripts/make_finals.py` does;
- writes it to a temporary folder in the submission format.

Result on 1 November 2026: **f2 SHA-256 `efbc9bcbe0d3386c…`, identical to the file Zindi scored (8DPgZvCX)**. The same holds for f5 (`5ff4d8370ab1ca34…`).

Depth 2: everything from the raw data, in a fresh copy.

```
python scripts/reproduce.py --full <empty folder>
```

The script:

1. copies the code to the folder and links the verified raw data;
2. creates a new virtual environment from `requirements.txt`;
3. runs the data check, the feature builders and the 24 scripts that wrote f2's signals, in their original order;
4. compares every regenerated signal file with the submitted one;
5. rebuilds f2 inside the copy.

Result on 1 November 2026, in one uninterrupted run from an empty folder (report in `docs/reproduction_report_2026-11-01.md`):

- the per-tract features rebuilt from the raw data equal the archived ones to within floating-point summation order (largest difference 1×10^{-8}, on footprint areas in m^2; DuckDB aggregates in parallel);
- **all 61 signal files of f2 regenerated bit for bit**, with the SHA-256 of the files Zindi scored;
- f2 and f5, rebuilt inside the copy, are identical to the submitted files;
- run time on the laptop above: **49 minutes**, made up of 4 minutes for the environment, 40 for the feature builders and 5 for the 24 signal scripts and the tests.

Two fixes were needed to get there. Both are part of the published code:

- **Older signals keep their original treatment of the water tracts.** The 46 signals prepared before 25 October predate the neutralisation of the two water tracts, so `bbx.signals.LEGACY_WATER_SCRIPTS` lets them keep their original values there.

- **The first blend sees the scores it saw at the time.** s16 (not part of f2) was built from the scores Zindi had returned by then, so the full mode records each score in the copy's log at its historical date.

Where things are:

Path	Content
`src/bbx/`	package: config, guarded I/O, features, structural model, signals, leaderboard algebra, blend, formula rule, submission writer
`scripts/make_*.py`	one script per submission batch, named after the files it writes
`scripts/make_finals.py`, `scripts/make_f5.py`	the final blends
`scripts/reproduce.py`, `scripts/make_doc_figures.py`	reproduction, and every figure and number of this document
`submissions/`	every submitted CSV, `log.csv` (SHA-256, purpose, expected and public RMSE, Zindi ID), `zindi_scores_final.csv` (public and private scores read after the close)
`docs/`	analysis, plan, decision log (`decisions.md`), daily journal (`journal.md`, French), audit plan (`05_plan_audit.md`), reproduction report (`reproduction_report_2026-11-01.md`)
`tests/`	unit and data-contract tests

Appendix A. Every submission

Expected = public RMSE logged before upload. Single signals were logged with a guessed correlation; blends were logged with the algebra of Section 6. ρ pub / priv = correlation with the truth implied by the public and by the private RMSE (Figure 2). The weight column gives the weight in f2, in units of the submitted file. f2 lists 61 signals; the eastern-wa indicator s31 is constant on the four scored regions and drops out. Files without a weight were left out of f2: blends, constants, the probe, and the WEAK signals (Section 6.5).

File	Zindi ID	What it measures	Expected	Public	Private	ρ pub / priv	Weight in f2
`s01_const_000`	hz2e3ofY	constant 0.00: measures target mean and sd	0.10817	0.10725	0.10757	—	
`s02_const_010`	f8rewtqK	constant 0.10: measures target mean and sd	0.09849	0.09768	0.09792	—	
`s03_strata_rural_tribal`	69ej3ZMW	strata-only prior: non-urban share, log land area, tribal share	0.08586	0.08254	0.08274	+0.38 / +0.39	-0.136
`s04_structural_v1`	eh4n4hWS	structural model v1: MC expectation of organiser formula (route proxy, physical fire ...	0.08339	0.08407	0.08591	+0.33 / +0.28	+0.193
`s05_fire_expected_gap`	FqGb62wz	expected fire-station gap from physical station density, misfiled names, Overture ...	0.08492	0.08889	0.08891	+0.15 / +0.17	-0.132
`s06_c_transport`	N6SChmQ9	structural v1 transport contribution (route proxy vs Overture scored classes)	0.08619	0.08210	0.08296	+0.42 / +0.40	-0.222
`s07_c_cbp`	hTMwh65B	structural v1 CBP-half contribution (places vs establishment proxy)	0.08801	0.08986	0.09104	+0.01 / -0.05	+3.111
`s08_c_school`	SB2WZozi	structural v1 school contribution	0.08858	0.08986	0.09047	-0.04 / -0.08	+1.232
`s09_region_ok`	iMRULWwD	regional offset: eastern-ok indicator	0.08619	0.08534	0.08625	+0.29 / +0.26	+0.082
`s10_region_az`	tjkmzErF	regional offset: maricopa-az indicator	0.08858	0.08868	0.08909	+0.09 / +0.08	+0.687
`s13_bld_msft_share`	NwoRLKoL	share of Overture buildings sourced from Microsoft ML	0.08619	0.08527	0.08540	+0.29 / +0.30	-0.017
`s14_rural_pop_no_fire`	JZp1WSeo	rural, populated tract with no Overture fire_department place	0.08722	0.08698	0.08691	+0.21 / +0.23	+0.110
`s16_blend_v1`	iu1dZ8Hf	blend of s03-s10 (ridge 0.05) weighted by leaderboard-implied covariances	0.07797	0.07820	0.07960	+0.48 / +0.45	
`s17_major_route_unscored_share`	Ntg2t2Pe	share of Interstate/US/state route km that Overture classes below secondary	0.08801	0.08905	0.08936	+0.07 / +0.07	-0.031
`s18_c_transport_tx`	18uR7Wqi	structural v1 transport contribution, south-central-tx only	0.08492	0.08836	0.08948	+0.18 / +0.14	+0.138
`s19_fire0_with_community`	3zQ1zL6n	no Overture fire_department place but a community anchor (post office, town hall, library...)	0.08801	0.08906	0.08923	+0.07 / +0.08	+0.196
`s20_partial_fire_gap`	M5HiaC4D	fewer Overture fire stations than the physical density prior expects	0.08874	0.08894	0.08922	+0.05 / +0.06	-0.059
`s22_no_ems_place`	YG6fcnXW	no Overture ambulance_and_ems_services place in the tract	0.08874	0.08931	0.09000	-0.00 / -0.05	+0.518
`s24_cbp_area_k1`	4udQPbF8	CBP half if establishments are allocated to tracts by land area (county-relative places ...	0.08874	0.08869	0.08924	+0.09 / +0.05	+0.114
`s25_cbp_pop_k05`	yy6VtcLw	CBP half if establishments are allocated by population, 2 places per establishment	0.08874	0.08995	0.09043	-0.09 / -0.11	-2.904
`s26_bld_per_hu_low`	tqeAsrZK	fewer Overture buildings than ACS housing units (1 - min(1, bld/hu))	0.08891	0.09001	0.09031	-0.20 / -0.20	-1.090

File	Zindi ID	What it measures	Expected	Public	Private	ρ pub / priv	Weight in f2
s27_bld_small_share	rzHKeus4	share of Overture footprints under 50 m2	0.08891	0.08863	0.08878	+0.11 / +0.15	-0.878
s28_bld_esri_share	uE8sY5kj	share of Overture footprints sourced from Esri Community Maps	0.08891	0.08914	0.08961	-0.00 / -0.04	-0.106
s29_poi_closed_share	1GcBxnfH	share of Overture places flagged permanently closed	0.08898	0.08942	0.08978	-0.14 / -0.15	+1.000
s30_import_share	LjUPFbH4	share of road km unedited since 2012 (TIGER-import era)	0.08874	0.08860	0.08910	+0.10 / +0.07	+0.039
s31_region_wa	BZoBWXPZ	regional offset: eastern-wa indicator	0.08891	0.08903	0.08901	—	
s32_region_ca	ZSQCCsi2	regional offset: northern-ca indicator	0.08900	0.08900	0.08934	+0.02 / +0.01	+0.055
s34_unnamed_share	9VvWPtRA	share of drivable road km without a primary name	0.08801	0.08721	0.08814	+0.21 / +0.16	-0.003
s36_cbp_hu_k2	K8jB5peB	CBP half if establishments are allocated by housing units, 0.5 place per establishment	0.08891	0.08962	0.09006	-0.11 / -0.14	+0.069
s37_logrel_places_per_bld	RJKHastf	log of places per building relative to the county (continuous)	0.08891	0.08924	0.08943	-0.02 / +0.01	+1.269
s38_cbp_area_k025	dStjCBas	CBP half if establishments are allocated by land area, 4 places per establishment	0.08891	0.08870	0.08914	+0.10 / +0.07	+0.913
s39_fire0_nbr300	hwyiUTTZ	no Overture fire_department place in the tract but one within ~300 m across its boundary	0.08874	0.08951	0.08986	-0.03 / -0.03	-0.068
s40_fire0_no_nbr1km	HtPC78m2	no Overture fire_department place in the tract nor within ~1 km of it	0.08858	0.08899	0.08911	+0.05 / +0.08	-0.004
s41_log_msft_area	fjQExKuP	mean footprint area of Microsoft-sourced Overture buildings (log)	0.08891	0.08868	0.08906	+0.10 / +0.09	-0.193
s42_zip_area_k05	sMEM8pzb	CBP half if ZIP establishments are spread to tracts by land area (2 places per establishment)	0.08891	0.08877	0.08925	+0.08 / +0.05	-0.161
s43_zip_area_k1	qnDMcmSv	CBP half if ZIP establishments are spread to tracts by land area (1 place per establishment)	0.08891	0.08898	0.08946	+0.04 / -0.00	+0.360
s77_t3_c_transport	DXzKv4BR	structural v3 transport contribution: boundary highways credited to both tracts, route …	0.07950	0.07426	0.07413	+0.56 / +0.57	-0.718
s78_t3_nomult_c_transport	MsRHPaKQ	structural v3 transport contribution without route multiplicity	0.08037	0.07553	0.07580	+0.54 / +0.54	+0.485
s79_t3_score	LFsjNG5t	structural v3 composite score	0.08159	0.07890	0.07999	+0.47 / +0.45	-0.849
s80_bgap_s300	WJLiKSsK	building gap contribution: hand-drawn footprints hiding max(area/300-1,0) Microsoft footprints	0.08858	0.09008	0.09039	-0.07 / -0.07	-0.110
s81_t3_tx	T6CXBsNH	structural v3 transport contribution (s77), south-central-tx only	0.07950	0.08817	0.08937	+0.25 / +0.22	+0.784
s82_t3_az	vLuZo1Wh	structural v3 transport contribution (s77), maricopa-az only	0.08801	0.08757	0.08740	+0.18 / +0.22	+0.771
s83_t3_ok	HfUp2qXG	structural v3 transport contribution (s77), eastern-ok only	0.08619	0.08153	0.08294	+0.44 / +0.40	+1.447
s84_t3_ca	b8u4KAU8	structural v3 transport contribution (s77), northern-ca only	0.08838	0.08809	0.08845	+0.15 / +0.14	+0.903
s86_cbp4_bld1000	F4wXJimX	CBP v4: business addresses proxied by bld_ge1000, gap 1-min(1,places/(k*proxy)), k=median	0.08858	0.08880	0.08960	+0.07 / +0.02	+0.500
s92_fire_osm_noov	YkHZBm4b	OSM fire-station building in the tract but no Overture fire_department place	0.08858	0.08808	0.08798	+0.16 / +0.20	+0.245

File	Zindi ID	What it measures	Expected	Public	Private	ρ pub / priv	Weight in f2
`s93_burn2016_x_dens`	vVi8P5tM	MTBS burned share (last fire 2016 or later) times log building density	0.08874	0.08883	0.08877	+0.07 / +0.12	-0.009
`s95_fire_misfiled_noov`	NtLg2JbE	name-verified fire station under another category, no Overture fire_department place	0.08858	0.08771	0.08817	+0.20 / +0.18	+0.907
`s97_t7_c_transport`	5Puciiag	structural v7 transport contribution: state loops/turnpikes as highways, TX FM/RM 0.112, OK …	0.07314	0.06648	0.06505	+0.67 / +0.70	+1.059
`s98_t7_p_defined`	ZSFx4Uph	structural v7 probability that the transport component is defined	0.08619	0.08490	0.08526	+0.31 / +0.30	-0.010
`s100_t8_c_fire`	ZLjAupNf	structural v8 fire contribution (clean fire-station evidence)	0.08858	0.08795	0.08824	+0.17 / +0.17	+2.036
`s101_t7_transport_gap`	u33fnwGi	structural v7 expected transport gap (component level)	0.07950	0.07883	0.07731	+0.46 / +0.50	-0.102
`s102_t7_p_transport_one`	17wt6cVc	structural v7 probability that the transport gap equals 1	0.08858	0.08885	0.08864	+0.07 / +0.13	-0.606
`s103_t8_p_fire_defined`	o5LTLYBQ	structural v8 probability that the reference has a fire station	0.08801	0.08700	0.08721	+0.22 / +0.23	+0.298
`s106_t9_c_transport`	PfWH69jt	structural v9 transport contribution: boundary-following routes (runs > 232 m) up to 30 m	0.07435	0.06902	0.06796	+0.64 / +0.66	+0.029
`s107_t8_c_fire_ok`	PAe91rog	structural v8 fire contribution (s100), eastern-ok only	0.08858	0.08734	0.08773	+0.23 / +0.23	+0.203
`s108_t8_c_fire_az`	VR4UNKc7	structural v8 fire contribution (s100), maricopa-az only	0.08886	0.08901	0.08954	+0.03 / -0.01	+1.125
`s109_t8_c_fire_ca`	SZcKAeBh	structural v8 fire contribution (s100), northern-ca only	0.08891	0.08882	0.08921	+0.07 / +0.06	+0.751
`s111_blend45_ridge0005`	EboKdJHP	validation blend of 45 measured signals (s07/s22 excluded), ridge 0.005, formula rules	0.05974	0.05999	0.06126	+0.74 / +0.73	
`s112_t7_p_one_tx`	ANjh6SFV	structural v7 probability that the transport gap equals 1 (s102), south-central-tx only	0.08891	0.08906	0.08908	+0.02 / +0.08	
`s113_t7_p_one_ok`	LY13LjGG	structural v7 probability that the transport gap equals 1 (s102), eastern-ok only	0.08891	0.08880	0.08922	+0.07 / +0.05	
`s117_t8_p_score_ge30`	4xGiP53V	structural v8 probability that the composite score reaches 0.30	0.08492	0.08273	0.08261	+0.38 / +0.39	-0.563
`s119_ems_station_nofire`	H95w1uKD	public EMS station named in Overture, no fire_department place, weighted 1/(4 n_defined)	0.08874	0.08940	0.08940	-0.01 / +0.03	
`s124_wa_public_probe`	S7WAmtUM	probe: eastern-wa = 1, elsewhere m; RMSE = s (0.089022) if eastern-wa is not public, about …	0.08902	0.08902	0.23848	—	
`s125_t7_c_transport_houston`	JDbHXpTv	structural v7 transport contribution, counties 48201/48157/48339/48039/48167 only (1483 tracts)	0.08874	0.08869	0.08905	+0.09 / +0.08	
`s126_t7_c_transport_phoenix`	uEzkqmuj	structural v7 transport contribution, counties 04013 only (1009 tracts)	0.08874	0.08870	0.08901	+0.08 / +0.09	
`s127_t7_c_transport_sanantonio`	wZCBMqFh	structural v7 transport contribution, counties 48029/48091/48187 only (449 tracts)	0.08874	0.08862	0.08921	+0.10 / +0.06	
`s128_t7_c_transport_austin`	T6KGJMna	structural v7 transport contribution, counties 48453/48491/48209 only (471 tracts)	0.08874	0.08886	0.08907	+0.06 / +0.08	
`s133_t8_c_fire_bigmetros`	NPogZGD1	structural v8 fire contribution, large metro areas only (5762 tracts)	0.08891	0.08933	0.08984	-0.04 / -0.09	
`s134_t8_fire_towncore`	kTpYqDVb	v8 fire contribution with RUCA 4/7 town cores as station evidence, minus v8 (342 tracts)	0.08858	0.08799	0.08837	+0.17 / +0.16	+0.469

File	Zindi ID	What it measures	Expected	Public	Private	ρ pub / priv	Weight in f2
`s136_t8_fire_prior_recal`	zcWkiNzd	v8 fire contribution with 1 station per 2,000 rural and 36,000 urban residents, minus v8	0.08801	0.08631	0.08645	+0.28 / +0.29	+0.190
`s137_t9_p_score_ge30_delta`	427Fdgb3	P(score >= 0.30), v9 (town cores, rural/urban station rates) minus v8	0.08801	0.08611	0.08649	+0.29 / +0.29	
`s138_t9_p_zero_delta`	YJUXzQyn	P(score = 0), v8 minus v9 (town cores, rural/urban station rates)	0.08858	0.08773	0.08801	+0.19 / +0.20	
`s140_t7l_c_transport_delta`	NKYheKx3	v7 transport with local street names as extra TIGER records (weight 0.25), minus v7	0.08858	0.08630	0.08614	+0.35 / +0.40	+1.660
`s142_t10s_c_transport_delta`	ctyeEWrg	v7 transport with street-named FM (TX) and county routes (OK) as the S1200 ones, minus v7	0.08891	0.08895	0.08921	+0.04 / +0.05	
`s143_t9_fire_gapfill`	y4ftFjf6	v9 fire contribution with gap filling (rate x exp(0.35 z(log distance to nearest known ...	0.08858	0.08753	0.08764	+0.22 / +0.24	
`s146_t7lf_c_transport_delta`	hL3p3mJS	v7L plus Interstate and Texas loop local names (weight 0.25), minus v7L	0.08891	0.08851	0.08918	+0.14 / +0.06	+2.401
`s147_t7linv_c_transport_delta`	ZNPnDWMf	v7L with local names 0.35 and FM / OK county weights re-solved to the inverse ratio reading, ...	0.08858	0.08578	0.08650	+0.40 / +0.37	+0.602
`s150_t9lfi_p_score_ge30_delta`	6sUPkgD7	P(score >= 0.30) under v9LFinv (v9 fire, s146 + s147 transport) minus under v9	0.08880	0.08580	0.08640	+0.54 / +0.50	
`f2_blend_ridge0005_with_s07_s22`	8DPgZvCX	final blend of 61 measured signals, ridge 0.005, formula rules, four scored regions	0.05500	0.05751	0.05871	+0.76 / +0.75	
`f5_blend_ridge002`	LEGqiwj2	final blend, 58 signals (WEAK and s07/s22 out), ridge 0.02; sent after the close, not ranked	0.05675	0.05698	0.05805	+0.77 / +0.76	

Appendix B. Decision log (abridged from `docs/decisions.md`)

Date	Id	Decision
16 Oct	C1	The scoring pipeline never reads withdrawn layers, near-zero-RMSE scorecards or other participants' group means.
16 Oct	D1	The leaderboard measures at most 20 global parameters; no tract or small-group probing.
16 Oct	D2, D2-bis	Footprint-derived strata columns excluded, with a code guard.
18 Oct	D1-bis	Up to about 50 measured signals, continuous or groups of 300 or more tracts, ridge on every weight.
20 Oct	D2-ter	D2 confirmed until the staff answers #35402.
21 Oct	D1-ter	Up to 66 signals (principal components of unexplained permitted features allowed).
29 Oct	D1-quater	Up to 73 signals; finals drop signals without public gain (WEAK).
31 Oct	D1-quinquies	Up to 74 signals, for one last transport edge (s150, later found WEAK).